Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese	Department of Electrical, Electronic and Computer Engineering
Klastoets 1	**Class Test 1**
VRAESTEL	**QUESTION PAPER**
Kopiereg voorbehou	Copyright reserved
Module EBN111	**Module EBN111**
Elektrisiteit en Elektronika	**Electricity and Electronics**
February 2018	February 2018

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***	MEMO		
Maksimum punte: *Maximum marks:*	16	Volpunte: *Full marks:*	15
Duur van vraestel: *Duration of paper:*	30 min + 10 min to complete OCR form *30 min + 10 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		8 + OCR	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
   All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session.
Do step 1 in the first 30 minutes of the test
Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.
- The Assessment ID is: **2018EBN111C1**
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2														
04	-	1								Ω								

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie.
Doen stap 1 in die eerste 30 minute van die toets.
Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (-5 min to 0 min): Voltooi die voorblad van di OKH vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.
- Die "Assessment ID" is **2018EBN111C1**
- Gebruik 'n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerde gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2														
04	-	1								Ω								

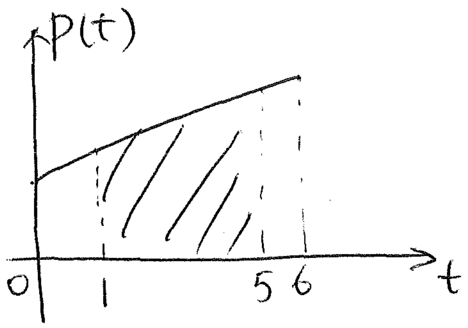
Questions 1-2 [4]	Vraag 1-2 [4]
A geyser is powered by 230 V and draws a current of 10 A for 90 minutes per day for water heating. If electricity is charged R 2 / kWh, how much does it cost (denoted by C) over 360 days (in Rand)?	'n Geyser word met 230 V aangedryf en trek 10 A vir 90 min per dag om water te verhit. Indien elektrisiteit R2/kWh kos, hoeveel kos dit (aangedui deur C) oor 360 dae? (in Rand)?
01 C	**01** C
The power consumption of a resistor increases linearly from 2 W at t = 0 s to 8 W at t = 6 s. Determine the energy (W) absorbed by the resistor between t = 1 s and t = 5 s. Hint: use graphic approach to solve the problem.	Die drywing wat deur 'n resistor verbruik word styg lineêr van 2 W by t = 0 s tot 8 W by t = 6 s. Bepaal die energy geabsorbeer deur die resistor tussen t = 1 s en t = 5 s. Wenk: Gebruik 'n grafiese benadering om die problem op te los.
02 W	**02** W

01: $C = 230 \times 10 / 1000 \times 1.5 \times 2 \times 360 = 2484$ Rand

02: $p(t) = t + 2$

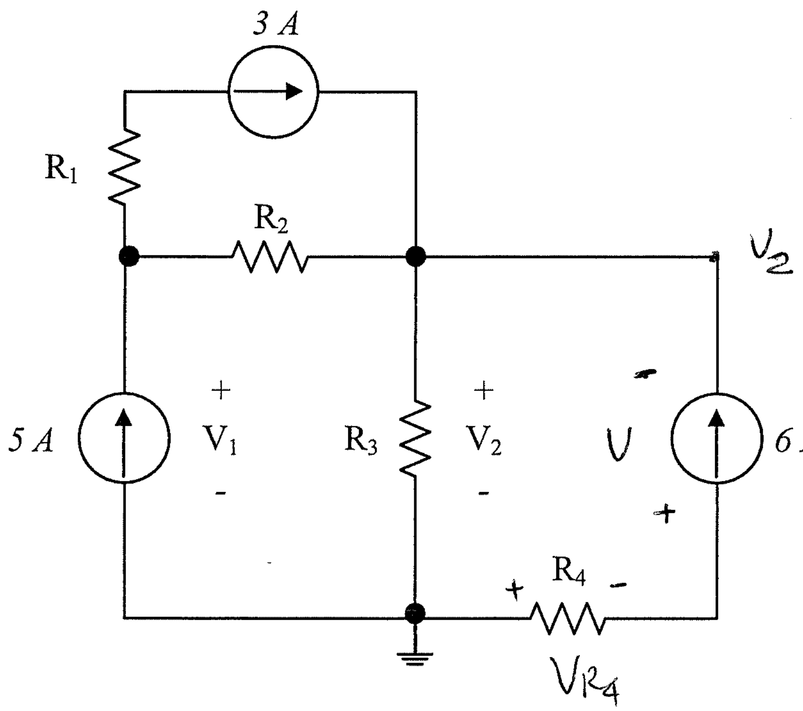
$p(1) = 3$

$p(5) = 7$



$W = \frac{1}{2}(3+7) \times (5-1) = 20\,J$

Questions 3-4 [2]		Vraag 3-4 [2]	
It is given that $V_1 = 50$ V, $V_2 = 40$ V, and $R_4 = 2\ \Omega$. Calculate the power associated with the *6 A* current source and the resistor R_2.		Dit word gegee dat $V_1 = 50$ V, $V_2 = 40$ V, and $R_4 = 2\ \Omega$. Bepaal die drywing geassosieer met die 6 A stroombron en die resistor R_2.	
03	P_{6A}	**03**	P_{6A}
04	P_{R2}	**04**	P_{R2}



04: $V_{R_2} = V_1 - V_2 = 10\,V$

$I_{R_2} = 5 - 3 = 2\,A$

$\Rightarrow P_{R_2} = V_{R_2} \cdot I_{R_2} = 20\,W$

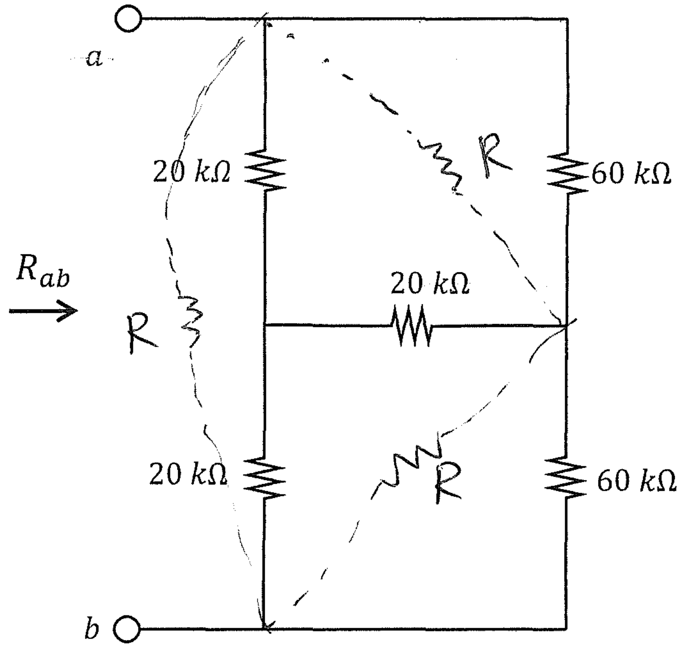
03: $V_2 = 40$, $V_2 + V + V_{R_4} = 0$

$V_{R_4} = 6 \times 2 = 12\,V$

$\Rightarrow V = -52\,V$

$\Rightarrow P_{6A} = -52 \times 6 = -312\,W$

Question 5 [2]		Vraag 5 [2]	
05	Determine R_{ab}.	05	Bepaal R_{ab}

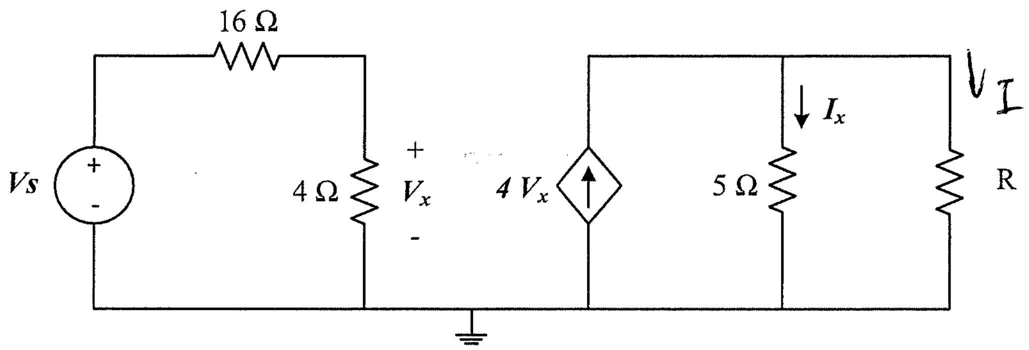


$R = 60\,k\Omega$

$R_{ab} = 60 \,||\, [\,60||60 + 60||60\,]$

$= 30\,k\Omega$

Questions 6-7 [4]		Vraag 6-7 [4]	
1) Determine V_s if I_x=3 A and R=5 Ω. 2) Determine R if V_s=10 V and I_x=3 A. (These are two independent questions.)		1) Bepaal V_s indien I_x = 3 A en R = 5 Ω. 2) Bepaal R indien V_s = 10 V en I_x = 3 A. (Hierdie is twee onafhanklike vrae.)	
06	V_s	**06**	V_s
07	R	**07**	R



1) $I_x = 3A, \quad R = 5\Omega \Rightarrow I = 3A$

$4V_x = 6 \Rightarrow V_x = 1.5V$

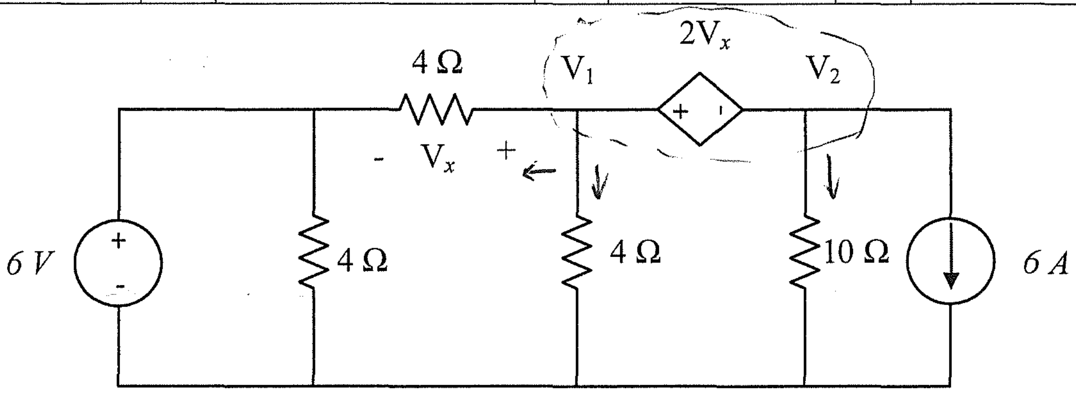
$V_x = V_s \cdot \frac{4}{20} = 1.5 \Rightarrow \underline{V_s = 7.5V.}$

2) $V_x = 10 \cdot \frac{4}{20} = 2V;$

$4V_x = 8$

$I_x = 3 = 8 \cdot \frac{R}{R+5} \Rightarrow \underline{R = 3\Omega}$

Questions 8 - 11 [4]				Vraag 8 - 11 [4]			
Use nodal analysis to set up two equations of the following form: $aV_1 + bV_2 = 12$ $cV_1 + dV_2 = -150$ Give the coefficients *a*, *b*, *c* and *d*.				Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $aV_1 + bV_2 = 12$ $cV_1 + dV_2 = -150$ Gee die koeffisiente *a*, *b*, *c* en *d*.			
08	*a*	**09**	*b*	**08**	*a*	**09**	*b*
10	*c*	**11**	*d*	**10**	*c*	**11**	*d*



$$V_1 = 6 + V_x \Rightarrow V_x = V_1 - 6$$

$$V_1 - 2V_x = V_2 \Rightarrow V_1 - 2(V_1 - 6) = V_2$$

$$-V_1 + 12 = V_2 \Rightarrow V_1 + V_2 = 12$$

$\Rightarrow a = 1, \quad b = 1$

$$\frac{V_1 - 6}{4} + \frac{V_1}{4} + \frac{V_2}{10} + 6 = 0$$

~~$10V_1 + 2V_2 = -150$~~ 10V1+2V2=-90

~~$\Rightarrow c = 10; \quad d = 2$~~ 16.67V1+3.33V2=-150

c=16.67 d=3.33

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	8	E	B	N	1	1	1	C	1		

	Answer	Unit
01	$C =$ 2484	Rand
02	$W=$ 20	J
03	$P_{6A}=$ -312	W
04	$P_{R2}=$ 20	W
05	R_{ab} = 30	kΩ
06	$V_s=$ 7.5	V
07	$R=$ 3	Ω
08	$a=$ 1	*No units required*
09	$b=$ 1	
10	$c=$ 10	
11	$d=$ 2	